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Social Engineering: Hacking the Mind

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What is Social Engineering?

Social engineering = manipulating people to compromise security

Common tactics:

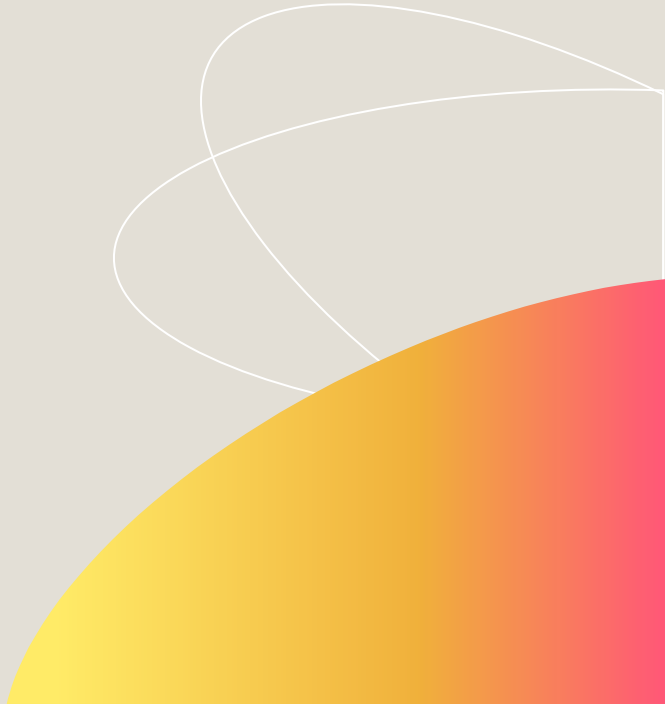
- Phishing
- Spear phishing
- Pretexting
- Vishing/Smishing
- Impersonation

Why Does Social Engineering Work?

Attackers exploit human psychology

Psychological triggers:

- Trust
- Fear
- Urgency
- Authority
- Curiosity
- Greed



A 4-Step Social Engineering Process

Step 1

Information Gathering

Learn as much as possible about the target to find useful weaknesses.

Step 2

Relationship Development

Build trust and establish credibility with the target.

Step 3

Exploitation

Manipulate the target into providing information, access, or taking a desired action.

Step 4

Execution

Carry out the attack using the information or access obtained.

Goal

Finally, achieving the attacker's ultimate objective, such as stealing information, money, credentials or gaining unauthorized access.

Real World Case Studies

Case	Social engineering tactic
Bangladesh Bank	Phishing + fake job applicant
BitPaymer	Spear phishing + malicious resume
W-2 Scam	Executive impersonation

The Social Side of Cybersecurity

Cybersecurity professionals need more than technical skills

Systemic thinking

Teamwork

Technical + Social skills

Civic duty

Continued learning

communication

Year/Quarter/Month

How Can we Reduce the Risk?

People

- Security awareness training
- Phishing simulations
- Verify unusual requests
- Report suspicious activity
- Slow down before acting

Technology

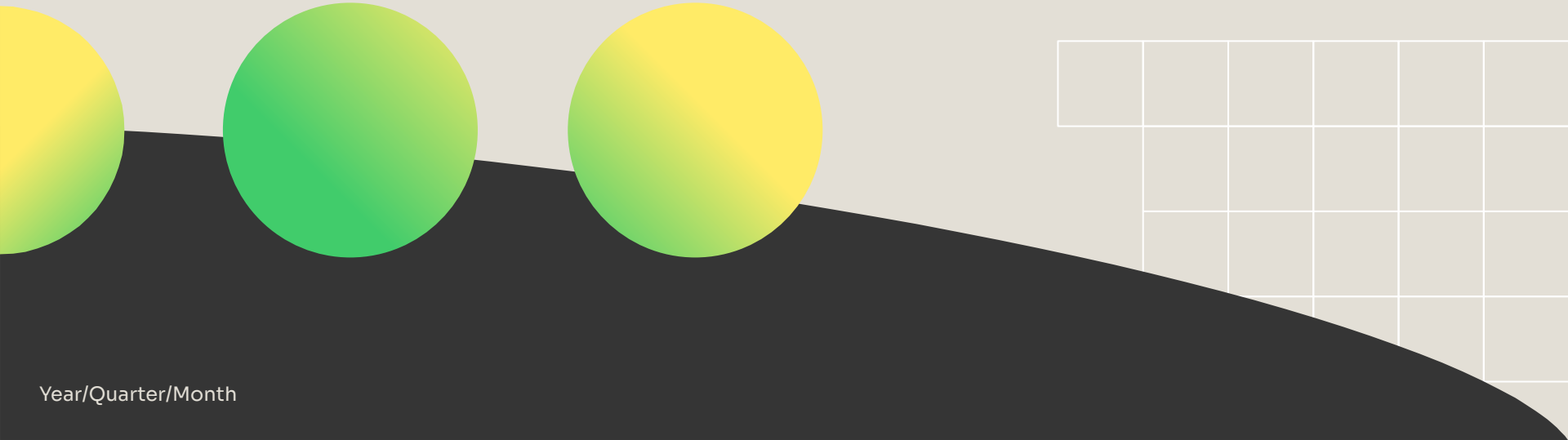
- MFA
- Email filtering
- Endpoint protection
- Security monitoring



Conclusion

The Human Element Matters

Key takeaway: Attackers may target technology but social engineering targets people




Thank you

References

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